

# BubbleID-Flow: Reproducible Machine-Vision Quantification of Projected Vapor Coverage in Subcooled Flow Boiling

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## Abstract

Quantitative flow-boiling imaging requires segmentation accuracy, temporal dependence, and thermal provenance to be evaluated together. BubbleID-Flow applies one-class Detectron2 Mask R-CNN to 3000-fps images of subcooled FC-72 flow boiling in a 2.5 mm by 5.0 mm channel with a 114.6 mm heated length. The model was fine-tuned from 130 annotated images (104 training; 26 same-sequence holdout). On the holdout, its union mask achieved mean intersection over union of 0.652, Dice coefficient of 0.769, and projected-coverage mean absolute error of 0.0051, compared with 0.1891 for Otsu morphology and 0.0327 for a training-only Gaussian pixel classifier. The split is not independent: 25 of 26 holdout images have a training image within one source-sequence index, precluding experiment-level generalization and individual-bubble claims. Four complete 45 V sequences contained 88–108 frames. Circular moving-block bootstrap intervals were 2.0–3.6 times wider than independent-frame Student- $t$  intervals; effective sample sizes were 3.5–11.2. Recomputed sequence means differed from archived summaries by 0.00085 on average. Across 36 thermally matched states, projected coverage correlated with electrical heat flux within each case ( $\rho_s = 0.893$ –1.000), but voltage and heat flux were perfectly rank-confounded. Adding coverage to a thermal baseline of heat flux, mass flux, and inlet subcooling did not

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improve leave-one-case-out prediction of seven-sensor surface temperature (RMSE: 6.92 versus 6.54 degC). The workflow therefore supports auditable two-dimensional projected coverage, not volumetric void fraction, universal regime criteria, or independently validated heat-transfer prediction.

*Keywords:* Flow boiling, Computer vision, Projected vapor coverage, Mask R-CNN, FC-72, Temporal uncertainty, Reproducibility

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## 1. Introduction

Flow boiling supports high heat removal in compact thermal systems, but its performance depends on mass flux, inlet subcooling, geometry, heating configuration, and evolving liquid–vapor structure [1–4]. Thermal measurements provide heat input and surface response, whereas high-speed images expose projected interfacial structure. Relating these streams is attractive for heat-transfer diagnosis, but an image-derived quantity is useful only if its definition, segmentation error, sampling uncertainty, and relationship to the thermal measurements are explicit.

Recent methods have moved boiling-image analysis from manual descriptors toward learned segmentation and prediction. U-Net and deep-learning workflows have segmented subcooled-boiling bubbles and their dynamics [5, 6]. BubbleMask and related methods have connected visual features to boiling and critical-heat-flux prediction [7, 8], while BubbleID provides an open framework for interface segmentation and tracking [9]. Image-derived features have also been used for heat-flux estimation and thermal-field reconstruction [10–12]. Most directly relevant to the present facility, Naderi et al. analyzed 107 mini-channel flow-boiling cases and more than  $3.8 \times 10^5$  images to connect spatial interfacial descriptors with heat-transfer prediction [13]. That larger study establishes the appropriate scale for predictive generalization. The narrower need addressed here is an auditable small-data workflow that exposes when a segmentation result is only a same-sequence reconstruction and when an apparent optical–thermal trend is explained by the imposed operating sweep.

Three questions motivate this work. First, does the archived BubbleID-Flow checkpoint improve projected-coverage reconstruction relative to simple and learned pixel baselines? Second, how much does serial dependence among high-speed frames inflate uncertainty in a state mean? Third, does projected coverage add cross-condition information about the measured surface-

30 temperature response beyond electrical heat flux, mass flux, and inlet sub-  
 31 cooling? The resulting contribution is a reproducibility and claim-boundary  
 32 study rather than a new heat-transfer correlation. Specifically, the paper (i)  
 33 documents the annotation, model, inference, and combined-mask calculation;  
 34 (ii) evaluates operating-point sensitivity, split leakage, and two baselines; (iii)  
 35 replaces independent-frame confidence intervals with a moving-block boot-  
 36 strap; and (iv) audits the thermal source before testing incremental optical  
 37 information. Acoustic-emission files are present in the project archive but  
 38 are excluded from quantitative analysis because trigger alignment and sen-  
 39 sor coupling have not been verified.

## 40 **2. Experimental data**

### 41 *2.1. Flow loop and test section*

42 The data were collected during CWRU Test 17 on October 17, 2025. The  
 43 working fluid was FC-72. The rectangular flow passage was 2.5 mm wide and  
 44 5.0 mm high, giving a cross-sectional area of  $1.25 \times 10^{-5} \text{ m}^2$  and a hydraulic  
 45 diameter of 3.33 mm. The bottom wall was electrically heated over a length  
 46 of 114.6 mm, corresponding to a nominal heated area of  $2.865 \times 10^{-4} \text{ m}^2$ .  
 47 Seven test-section surface-temperature locations were recorded at

$$x = [0.0054, 0.0227, 0.0400, 0.0573, 0.0746, 0.0919, 0.1092] \text{ m}$$

48 from the upstream end of the heated length. Figure 1 shows the loop, mea-  
 49 surement streams, heated test section, and test matrix.

50 The four image cases have nominal mass flow rates of 5, 10, 15, and 25 g/s  
 51 and nominal inlet subcoolings of 22, 22, 20, and 20 degC, respectively. With  
 52 the measured flow area, the matched thermal records give mean mass fluxes  
 53 of approximately 395–443, 781–824, 1189–1209, and 1994–2010  $\text{kg m}^{-2} \text{ s}^{-1}$ .  
 54 Across the matched states, electrical heat flux spans 6.51–41.76  $\text{W cm}^{-2}$ .  
 55 Table 1 identifies the available optical and thermal states.

56 Folder names combine applied voltage with operator labels such as onset  
 57 or CHF. No objective onset, wall-dryout, or critical-heat-flux criterion was  
 58 found in the supplied test notes. These labels are therefore retained only as  
 59 archive identifiers. In the analysis and conclusions, the last folder in a sweep  
 60 is called the operator-labeled endpoint; it is not treated as verified CHF.

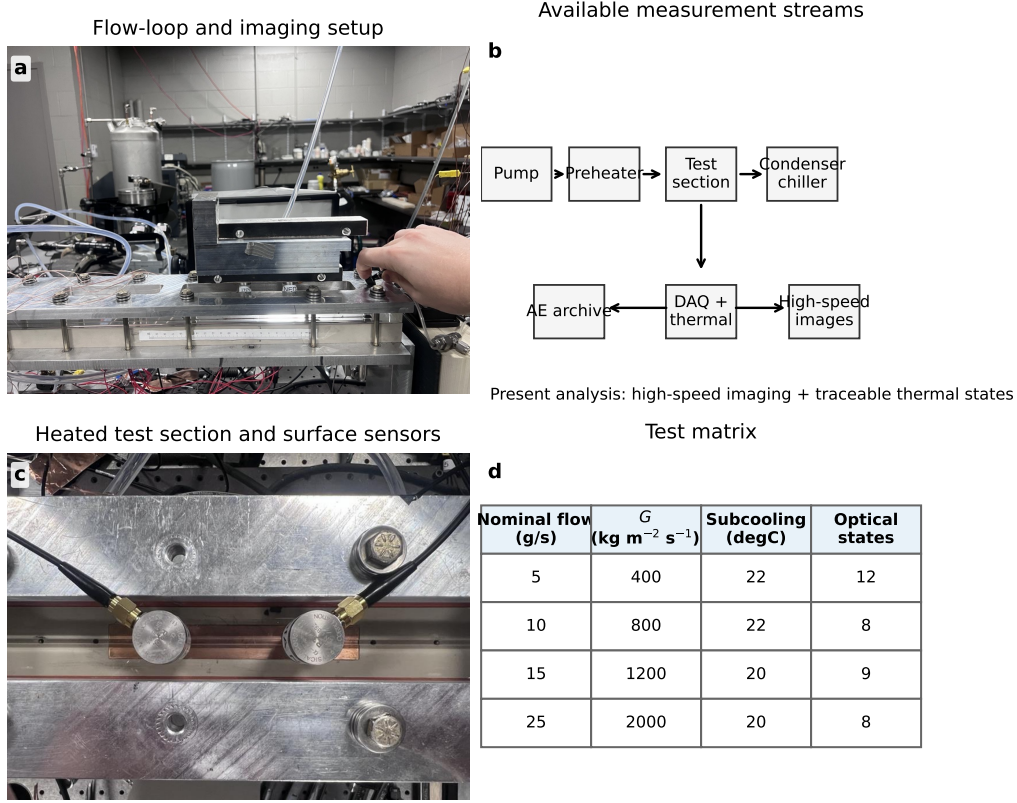


Figure 1: Facility and available data. (a) Flow-loop and imaging setup. (b) Loop and acquisition streams; the acoustic-emission archive is not analyzed because timing and coupling verification are unavailable. (c) Heated test section and surface-sensor access. (d) Nominal test matrix.

Table 1: Operating cases and auditable state coverage. One 10 g/s optical state at 45 V has no thermal row within the matching tolerance.

| Case     | $\dot{m}$ (g/s) | $G$ ( $\text{kg m}^{-2} \text{s}^{-1}$ ) | Optical states | Matched states |
|----------|-----------------|------------------------------------------|----------------|----------------|
| 5gs_22C  | 5               | 395–443                                  | 12             | 12             |
| 10gs_22C | 10              | 781–824                                  | 8              | 7              |
| 15gs_20C | 15              | 1189–1209                                | 9              | 9              |
| 25gs_20C | 25              | 1994–2010                                | 8              | 8              |

## 2.2. Optical and thermal records

A FASTCAM SA-Z recorded 1024 by 1024 pixel images at 3000 frames/s. The model analyzes a fixed 1024 by 70 pixel near-wall region of interest (ROI) starting at image coordinate (0, 485). The pixel-to-length calibration and spatial registration between the ROI, heated surface, and seven temperature locations were not supplied. Accordingly, this paper reports neither dimensional bubble size nor local optical-thermal coupling.

The thermal workbooks contain time, mass flow, voltage, electrical power, inlet subcooling, inlet and outlet temperature, pressure, and seven test-section surface-temperature streams. Instrument model numbers, calibration records, heat-loss characterization, and sensor uncertainty were not available. Scatter within a selected time window is consequently reported as temporal variability, not as measurement uncertainty.

The archive also contains two-channel EasyAE acoustic-emission (AE) records: event-level HIT exports, 10 ms continuous TIME exports, and binary waveform-stream (WFS) files. The event and continuous streams use a 38 dBAE threshold. Archive notes place the sensors approximately at 25% and 75% of the heated length, but also identify mounting adequacy as a concern. These files enable a limited multimodal screening below; they do not provide a verified common trigger with the thermal or image acquisitions.

## 3. Methods

### 3.1. Annotation, fine-tuning, and inference

The annotation dataset contains 130 images and 130 Labelme JSON files. The labels `bubble` and `bubble_cluster` are intentionally merged into one vapor-region category because the archived checkpoint is not a validated morphology classifier. Every fifth sorted image was assigned to the holdout, giving 104 training images with 7273 retained ROI annotations and 26 hold-out images with 1908 annotations. Figure 2 summarizes the conversion to one-class COCO data.

The Detectron2 Mask R-CNN model used a ResNet-50 feature-pyramid backbone [14, 15]. The archived configuration specifies 2000 iterations, evaluation every 500 iterations, one image per batch, a base learning rate of  $2.5 \times 10^{-4}$ , and anchor sizes of 8, 16, 32, 64, and 128 pixels. Figure 3 shows the retained model-artifact flow. The original training log, seed, augmentation trace, and checkpoint-selection record were not supplied, so this study evaluates the archived weights but does not claim to reproduce fine-tuning.

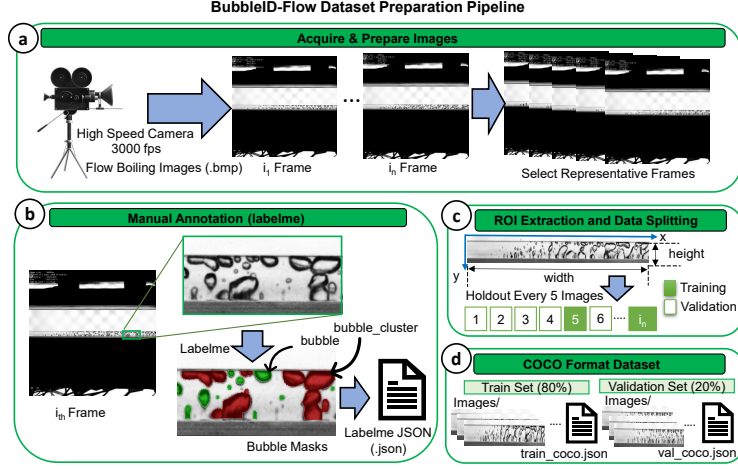


Figure 2: Dataset preparation from high-speed frames and Labelme polygons to cropped, one-class COCO training and same-sequence holdout sets. This project-team schematic is used directly from the updated model package.

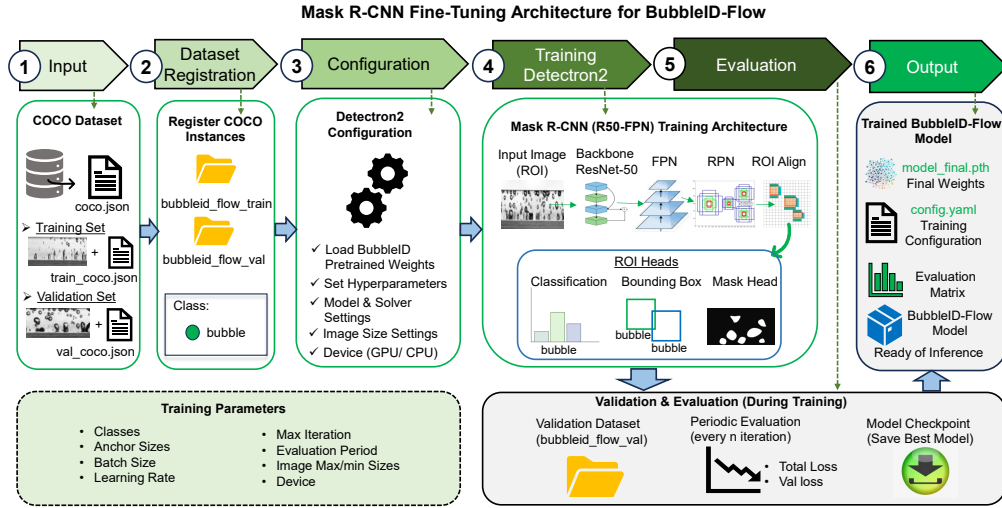


Figure 3: Archived Mask R-CNN fine-tuning and inference workflow. This project-team schematic is used directly from the updated model package.

97 The archived inference workflow prescribed a score threshold of 0.30 and  
 98 at most 300 detections per image. All retained instance masks are combined  
 99 before calculating projected vapor coverage,

$$\alpha_A = \frac{N_{\text{vapor}}}{N_{\text{ROI}}}, \quad (1)$$

100 where  $N_{\text{vapor}}$  is the number of pixels in the union mask and  $N_{\text{ROI}}$  is the ROI  
 101 pixel count. Thus  $\alpha_A$  is a two-dimensional projected area fraction in the  
 102 camera plane, not a volumetric void fraction.

103 Standard COCO metrics evaluate instance masks at a 0.05 score thresh-  
 104 old. The engineering endpoint in Eq. (1) is evaluated at the archived 0.30  
 105 setting using union-mask IoU, Dice coefficient, signed coverage error, and  
 106 absolute coverage error. A post hoc score-threshold sweep from 0.05 to 0.70  
 107 and detection-cap sweep from 50 to 300 assess sensitivity; they are not used  
 108 as independent tuning evidence. Percentile intervals for image-level mean  
 109 metrics use 10,000 deterministic bootstrap resamples. The split audit parses  
 110 source-frame indices and computes nearest training–holdout proximity. Be-  
 111 cause all annotations derive from one source sequence, the holdout evaluates  
 112 same-sequence reconstruction only.

### 113 3.2. Segmentation baselines

114 Two baselines test whether the deep model provides meaningful improve-  
 115 ment. The fixed baseline applies grayscale Otsu thresholding followed by  
 116 morphological cleanup, with no holdout tuning. The learned pixel baseline  
 117 models vapor and background intensities with diagonal Gaussian likelihoods.  
 118 Seventy-seven training images fit the class distributions; 27 sequence-grouped  
 119 training images form a calibration subset for the probability threshold. Nei-  
 120 ther baseline uses the 26-image holdout for parameter selection. The same  
 121 union-mask metrics are then evaluated on the holdout for all three methods.

### 122 3.3. Temporal uncertainty

123 Every available frame in the 45 V folder is processed for each case. Serial  
 124 dependence is summarized by the integrated autocorrelation time

$$\tau_{\text{int}} = 1 + 2 \sum_{k=1}^K \rho_k, \quad n_{\text{eff}} = \frac{n}{\tau_{\text{int}}}, \quad (2)$$

where the sum stops at the first non-positive autocorrelation in the initial positive sequence. Mean confidence intervals use a circular moving-block bootstrap with block length  $\lceil \tau_{\text{int}} \rceil$ , 5000 resamples, and a fixed random seed [16, 17]. Independent-frame Student- $t$  intervals are retained only as a diagnostic of underestimation.

### 3.4. Traceable thermal reduction

Only quantities recoverable from supplied measurements and documented geometry are used. Electrical heat flux and mass flux are

$$q_e'' = \frac{P_e}{WL_h}, \quad G = \frac{\dot{m}}{WH}, \quad (3)$$

with  $W = 2.5$  mm,  $H = 5.0$  mm, and  $L_h = 114.6$  mm. The reconstructed  $q_e''$  agrees with the workbook heat-flux column to numerical precision. For each voltage-labeled state, rows within  $\pm 0.75$  V are separated into contiguous blocks using a 5 s gap criterion; the largest non-overlapping block is retained. State means and standard deviations are then calculated for power, heat flux, mass flow, inlet and outlet temperatures, and the seven surface-temperature streams.

The source audit did not support use of the supplied heat-transfer coefficient, vapor quality, friction factor, or energy-balance columns. The archived reduction includes an undocumented multiplicative correction and unresolved unit and branch-logic ambiguities, while calibration, heat-loss, property-correlation, and uncertainty records are missing. Those derived quantities are therefore excluded rather than repaired through manuscript wording.

### 3.5. Optical-thermal association test

Spearman correlations between  $q_e''$  and  $\alpha_A$  describe each voltage sweep. Because applied voltage and heat flux are monotonic together, these correlations do not isolate an optical effect. Incremental information is tested with ridge regression for the mean of the seven measured surface temperatures,  $\overline{T}_s$ . The thermal baseline uses standardized  $q_e''$ ,  $G$ , and inlet subcooling; the augmented model adds  $\alpha_A$ . Ridge strength is fixed at 1.0. Performance is calculated by leave-one-state-out and the stricter leave-one-case-out cross-validation. All preprocessing is fit within each training fold.

### 155 3.6. *Exploratory timestamp-registered acoustic screening*

156 The 10 g/s case was selected for a bounded acoustic screen because its  
157 seven thermally matched states have optical summaries, the AE HIT and  
158 TIME records span the thermal run, and the WFS stream covers the run.  
159 The thermal record started at 11:41:47.605 and the AE record at 11:42:13.000  
160 on October 17, 2025, giving an archive timestamp offset of 25.395 s. Each  
161 thermal state interval was shifted by this offset to select an AE-relative win-  
162 dow. This is timestamp registration, not a common hardware trigger or a  
163 synchronization validation.

164 For each registered interval and AE channel, we report vendor-defined  
165 continuous average signal level (ASL), HIT rate, and median HIT amplitude.  
166 Absolute-energy fields are retained only as vendor units in the reproducibility  
167 tables and are not interpreted as calibrated acoustic energy. The binary WFS  
168 file is decoded with the `decode-wfs` message format and count-to-voltage  
169 calibration using a bounded message-by-message reader [18]; a 512-record  
170 prefix verifies two 1 MHz channels but is not state matched. The reusable  
171 parser and state-window implementation are retained in AELab [19]. No  
172 acoustic regression, source localization, or causal optical–acoustic analysis is  
173 performed.

## 174 4. Results and discussion

### 175 4.1. *Checkpoint reconstruction and baseline comparison*

176 The same-sequence holdout gives bounding-box  $AP_{50} = 0.516$  and mask  
177  $AP_{50} = 0.521$ ; AP averaged over IoU 0.50–0.95 is 0.234 for boxes and 0.249  
178 for masks. At the archived 0.30 operating threshold, the combined masks give  
179 mean IoU 0.652, mean Dice 0.769, coverage MAE 0.0051, and mean signed  
180 error +0.0021. Bootstrap 95% intervals are 0.573–0.721 for mean IoU, 0.690–  
181 0.830 for mean Dice, and 0.0037–0.0066 for mean absolute coverage error.  
182 Figure 4 shows representative predictions.

183 The error is coverage-dependent. Mean IoU is 0.444 for nine low-coverage  
184 images ( $\alpha_A \leq 0.05$ ), 0.758 for 13 intermediate images, and 0.775 for four  
185 high-coverage images. Low-coverage images have mean signed error +0.0072,  
186 whereas the high-coverage group has mean signed error  $-0.0054$ . Figure 5  
187 also shows that overlap metrics peak at a different post hoc threshold than  
188 coverage MAE. A single operating point is therefore not a universal instance-  
189 segmentation optimum.

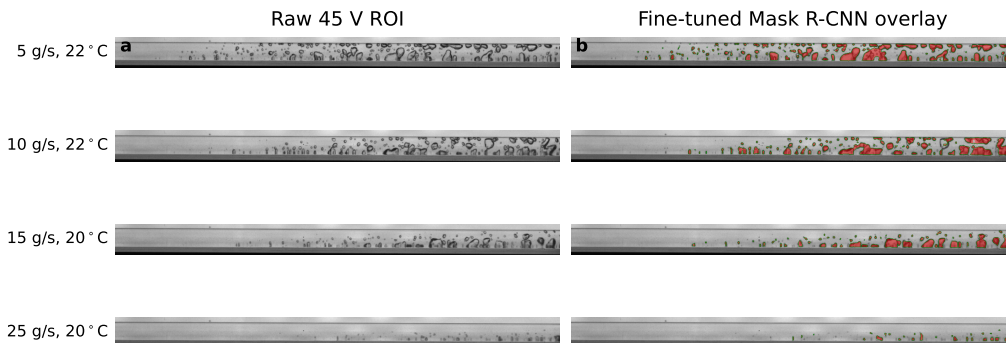


Figure 4: Representative 45 V near-wall ROIs for the four cases. Top: raw ROIs. Bottom: union-mask overlays from the archived one-class Mask R-CNN checkpoint at score threshold 0.30 and a 300-detection cap.

190 Table 2 and Fig. 6 show that Mask R-CNN substantially outperforms both  
 191 baselines for coverage reconstruction. The Gaussian pixel classifier improves  
 192 greatly over fixed Otsu morphology but still has 6.4 times the Mask R-CNN  
 193 coverage MAE. This comparison supports the learned instance model for the  
 194 stated union-coverage endpoint, while the modest instance AP continues to  
 195 preclude reliable bubble counts and sizes.

Table 2: Holdout segmentation comparison. The two baselines do not use holdout labels for parameter selection.

| Method          | Mean IoU | Mean Dice | Coverage MAE | Signed error |
|-----------------|----------|-----------|--------------|--------------|
| Otsu morphology | 0.209    | 0.331     | 0.1891       | +0.1891      |
| Pixel Gaussian  | 0.371    | 0.529     | 0.0327       | +0.0258      |
| Mask R-CNN      | 0.652    | 0.769     | 0.0051       | +0.0021      |

196 The apparent strength of the aggregate metrics must be bounded by the  
 197 split audit: ten holdout images share a nominal source-frame index with a  
 198 training image through filename variants, and 25 of 26 have a training image  
 199 within one source-sequence index. The evaluation therefore demonstrates  
 200 same-sequence reconstruction, not robustness to a new run, camera, fluid,  
 201 facility, or operating path.

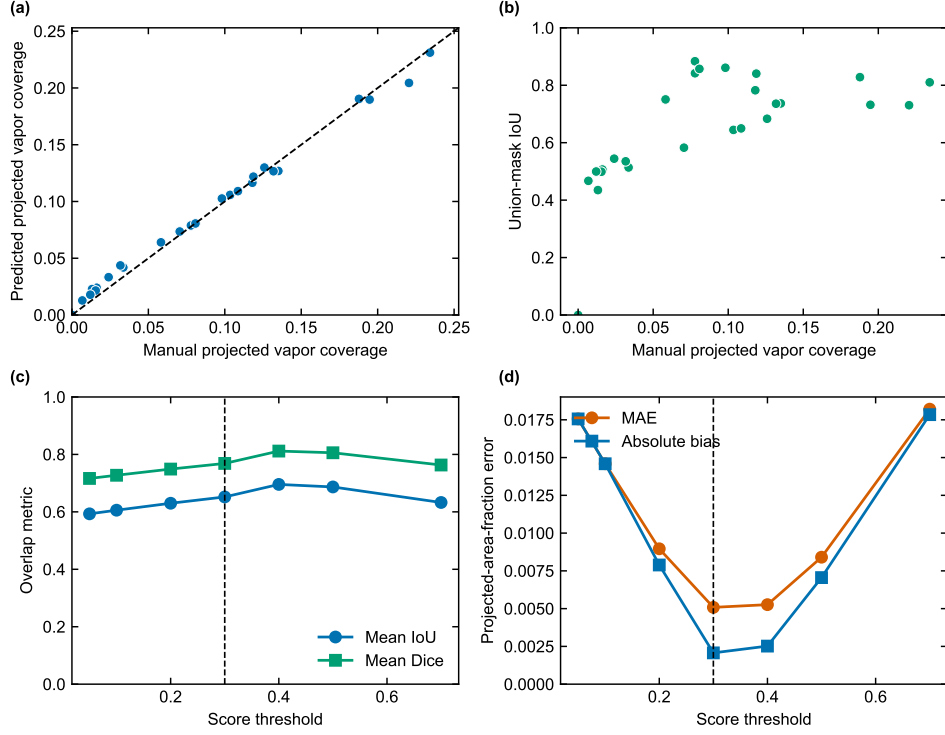


Figure 5: Same-sequence holdout sensitivity. (a) Predicted versus manual projected coverage. (b) Union-mask IoU versus manual coverage. (c) Mean IoU and Dice over score thresholds. (d) Coverage error over thresholds. Dashed vertical lines mark the archived 0.30 operating setting; the sweep is post hoc sensitivity analysis, not independent model selection.

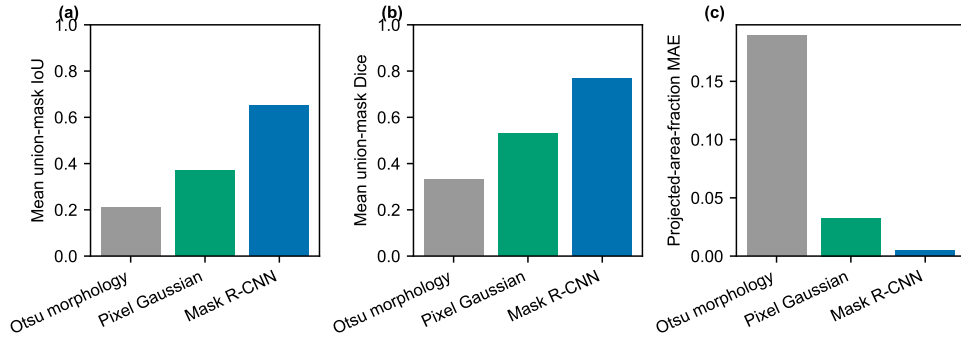


Figure 6: Simple and learned baseline evaluation on the 26-image holdout. (a) Predicted versus manual coverage. (b) Mean union-mask IoU and Dice. (c) Mean absolute coverage error. The Gaussian classifier is fit and calibrated using training images only.

#### 202 4.2. State trends and autocorrelation-aware uncertainty

203 The archived 37-state summaries span  $\alpha_A = 0$ –0.247. The 5 and 15 g/s  
 204 sweeps increase to 0.247 and 0.190, respectively, over their recorded voltage  
 205 folders. The 10 g/s sweep reaches 0.197 at 52.5 V and decreases to 0.076 in  
 206 the final 55 V folder; the 25 g/s sweep reaches 0.113 at 57.5 V and decreases  
 207 to 0.094 in the final 59 V folder. Because these endpoint folders lack an  
 208 objective CHF criterion and most states lack regenerated frame-level outputs,  
 209 the decreases cannot establish dryout or a universal transition threshold.

210 The four complete 45 V sequences contain 100, 108, 88, and 95 frames.  
 211 Their recomputed means are 0.12897, 0.10882, 0.06556, and 0.01378, com-  
 212 pared with archived values of 0.13086, 0.10966, 0.06536, and 0.01423. The  
 213 four-case mean absolute difference is 0.00085. Table 3 shows why treat-  
 214 ing frames as independent is inappropriate. Moving-block half-widths are  
 215 0.00103–0.00635, compared with 0.00044–0.00176 from Student- $t$  intervals.

Table 3: Autocorrelation-aware uncertainty for complete 45 V sequences. CI denotes the 95% moving-block-bootstrap interval.

| Case   | Frames | $\tau_{\text{int}}$ | $n_{\text{eff}}$ | Mean $\alpha_A$ | 95% CI          |
|--------|--------|---------------------|------------------|-----------------|-----------------|
| 5 g/s  | 100    | 10.19               | 9.82             | 0.12897         | 0.12631–0.13190 |
| 10 g/s | 108    | 9.64                | 11.20            | 0.10882         | 0.10514–0.11255 |
| 15 g/s | 88     | 25.00               | 3.52             | 0.06556         | 0.05905–0.07175 |
| 25 g/s | 95     | 9.84                | 9.66             | 0.01378         | 0.01274–0.01481 |

216 Agreement at 45 V verifies the documented image-to-coverage implemen-  
 217 tation for a common state using the same checkpoint and source images. It  
 218 is not independent model validation. The effective sample sizes also show  
 219 that acquiring many adjacent frames does not provide the precision implied  
 220 by the raw frame count.

221 A checkpointed all-state rerun also completed the 25, 27.5, and 30 V  
 222 folders of the 5 g/s case before the CPU run was stopped. Their 88, 99,  
 223 and 103 frame means are 0.02195, 0.03251, and 0.04138, differing from the  
 224 archived summaries by  $-0.00115$ ,  $-0.00296$ , and  $-0.00234$ . These differences  
 225 are larger than two of the four 45 V differences and confirm that an archived  
 226 state summary is not interchangeable with a complete-sequence mean. The  
 227 partial three-state result is retained as provenance evidence but is not mixed  
 228 into the cross-case figure.

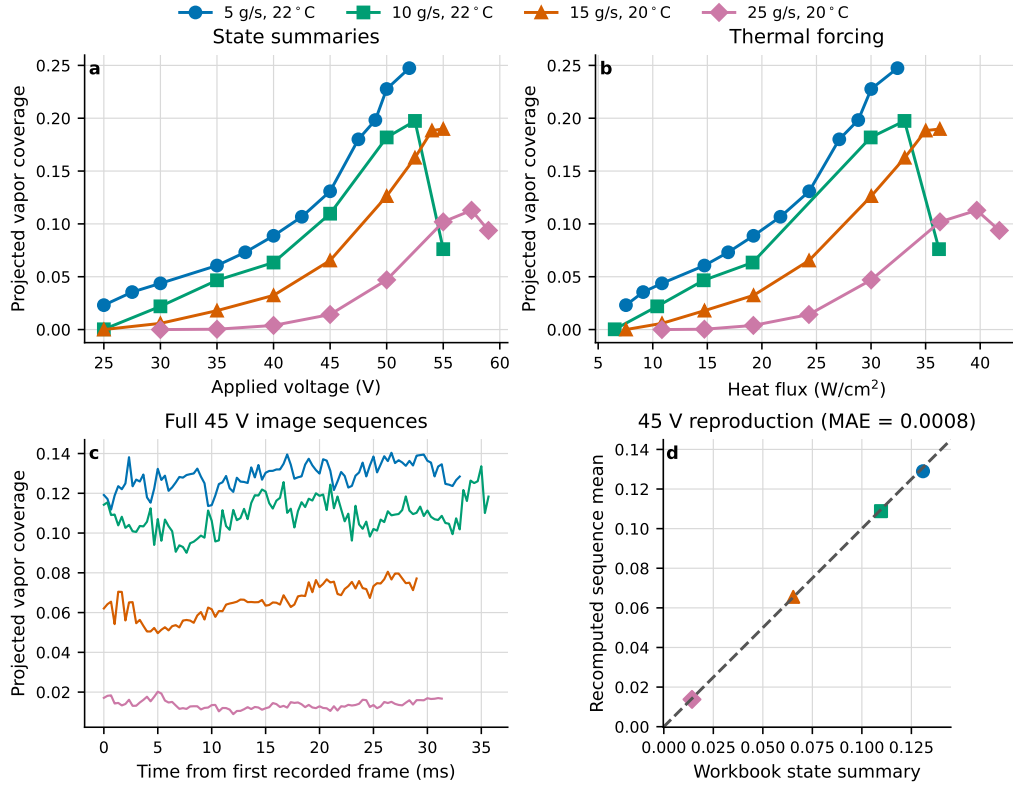


Figure 7: Optical state and sequence results. (a) Archived projected-coverage summaries versus voltage. (b) Summaries versus matched electrical heat flux; the 10 g/s, 45 V point is omitted because no thermal window matched. (c) Every available frame in the four complete 45 V sequences. (d) Recomputed sequence means versus archived 45 V summaries. Autocorrelation-aware numerical intervals are reported in Table 3.

### 229 4.3. Thermal association and incremental information

230 Within each voltage sweep, Spearman correlations between  $q_e''$  and  $\alpha_A$  are  
231 1.000, 0.893, 1.000, and 0.929 for increasing nominal mass flow. However,  
232 voltage and  $q_e''$  have  $\rho_s = 1.000$  in every case. The optical correlation is  
233 therefore inseparable from the imposed voltage progression in these data.

234 The cross-validated temperature test reinforces this limitation. For leave-  
235 one-state-out validation, the thermal baseline gives RMSE 4.84 degC and  
236 MAE 2.90 degC; adding  $\alpha_A$  worsens them to 5.57 and 3.11 degC. For leave-  
237 one-case-out validation, baseline RMSE and MAE are 6.54 and 5.45 degC,  
238 compared with 6.92 and 5.49 degC after adding coverage. Figure 8 shows the  
239 within-case correlations, confounding, and held-out predictions. The result  
240 does not imply that interfacial structure is thermally irrelevant. It shows  
241 that this projected ROI metric, sample size, and operating design do not  
242 demonstrate incremental predictive information beyond the three thermal-  
243 control variables.

### 244 4.4. Timestamp-registered multimodal screening

245 Figure 9 summarizes the seven 10 g/s states with archive timestamp reg-  
246 istration. Projected coverage increases from 0.00025 at 25 V to 0.197 at  
247 52.5 V, then decreases to 0.076 in the final 55 V folder. Over the same im-  
248 posed ramp, channel-1/channel-2 ASL increases from 24.38/23.38 dBAE to  
249 35.49/37.61 dBAE and the corresponding event rates increase from 1.92/2.12  
250  $\text{s}^{-1}$  to 928/638  $\text{s}^{-1}$ . The terminal optical decrease while AE measures remain  
251 elevated is a useful diagnostic that a single projected ROI metric and an AE  
252 activity metric are not interchangeable state indicators.

253 However, all three modalities are conditioned on the same voltage ramp,  
254 the acoustic windows depend on separate-system timestamps, and the archive  
255 reports possible sensor-coupling limitations. The figure therefore documents  
256 an exploratory multimodal co-variation screen, not a correlation coefficient,  
257 time-resolved coupling, acoustic origin, or a flow-regime classifier.

## 258 5. Reproducibility, limitations, and required validation

259 The repository records the checkpoint SHA256, ROI, score threshold, de-  
260 tection cap, source hashes, generated CSV files, and deterministic analysis  
261 settings. The relevant entry points are `scripts/evaluate_detectron2.`  
262 `py`, `scripts/evaluate_segmentation_baselines.py`, `scripts/prepare_`  
263 `aug9_optical_results.py`, `scripts/audit_thermal_reduction.py`, and

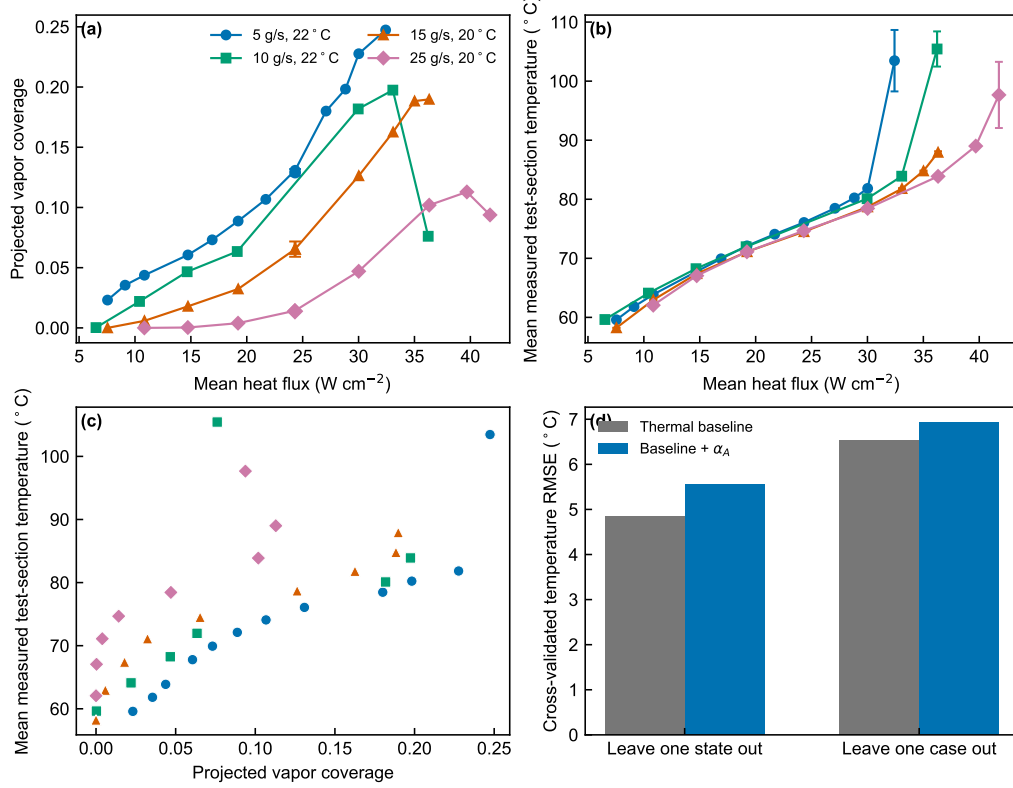


Figure 8: Optical-thermal association audit for 36 matched states. (a) Projected coverage versus electrical heat flux; moving-block intervals are overlaid at the available 45 V states. (b) Mean seven-sensor surface temperature versus heat flux; error bars show the temporal standard deviation of the instantaneous seven-location average within each matched window. (c) Mean surface temperature versus projected coverage. (d) Cross-validated temperature RMSE for the thermal baseline and the baseline augmented with projected coverage.

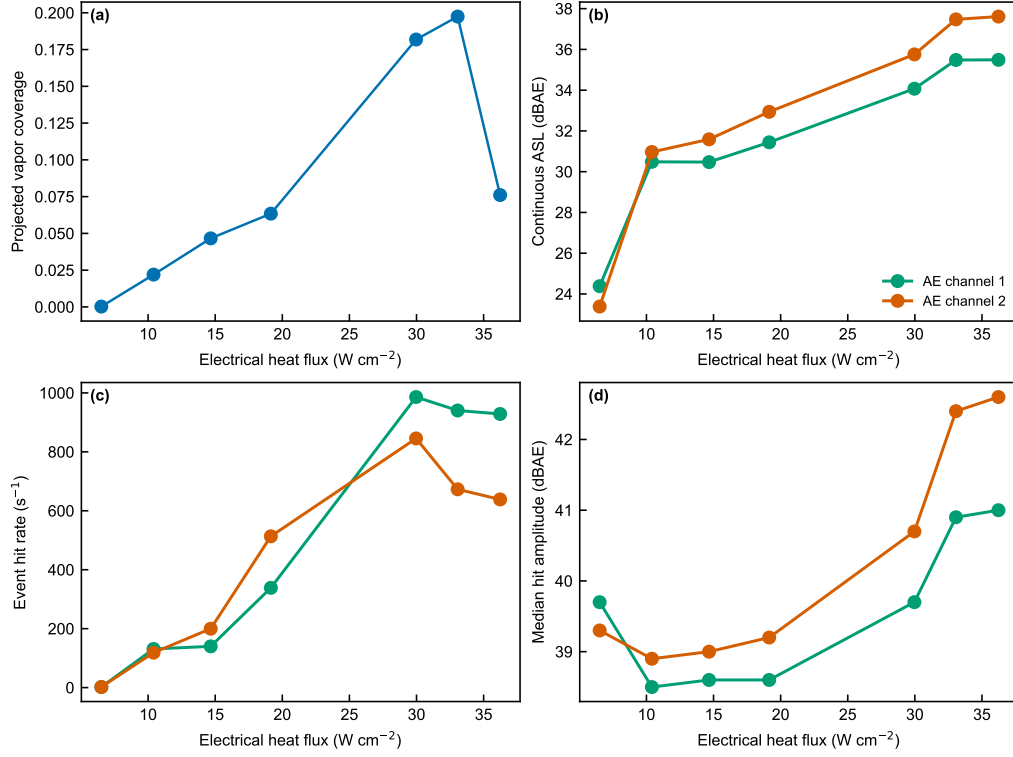


Figure 9: Exploratory 10 g/s multimodal state screen using archive timestamp registration. (a) Projected vapor coverage from the archived optical summary. (b) Continuous EasyAE average signal level. (c) Event hit rate. (d) Median event amplitude. Thermal state intervals are shifted by the 25.395 s difference between archive start timestamps. The records have no verified common trigger; lines show imposed-ramp co-variation only and do not establish optical–acoustic synchronization, causality, or source localization.

264 `scripts/analyze_optical_thermal_association.py`. The checkpoint  
 265 SHA256 is 6F60969CE876F57A78B53FC61895C99B92CB2E1B6F0E0B30F01F  
 266 A301236042CD.

267 Five limitations define the present claim ceiling. First, the manual hold-  
 268 out is strongly dependent on the training sequence, and the 0.30 operating  
 269 threshold has no experiment-independent calibration set. New manually an-  
 270 notated sequences held out by experiment and operating condition are re-  
 271 quired for generalization. Second, the four common 45 V folders and three  
 272 additional low-voltage 5 g/s folders have complete regenerated frame out-  
 273 puts; 30 state values remain traceable derived summaries without complete-  
 274 sequence temporal intervals. Third, the thermal records support electri-  
 275 cal heat flux, mass flux, temperatures, and their within-window variability,  
 276 but not a calibrated uncertainty budget or independently reproduced heat-  
 277 transfer coefficient, quality, friction, or energy balance. Fourth, the optical  
 278 ROI is not spatially registered to the heated surface or temperature loca-  
 279 tions and cannot measure three-dimensional void fraction. Fifth, the opera-  
 280 tor folder labels do not establish onset, CHF, or dryout. The 10 g/s acoustic  
 281 screen is limited to timestamp-registered state summaries; a common trig-  
 282 ger, quantified sensor coupling, calibration, and replicated runs are required  
 283 before any quantitative optical-acoustic synchronization or inference claim.

284 These are not resolved by weaker wording alone. The next validation ex-  
 285 periment should preserve trigger-synchronized image and thermal time bases;  
 286 record instrument models, calibration, heat loss, and uncertainty; define on-  
 287 set and CHF independently from the images; collect replicate ramps; and  
 288 reserve complete operating cases for annotation and blind evaluation. Until  
 289 those data exist, BubbleID-Flow should be used as a traceable projected-  
 290 coverage measurement within the present image domain.

## 291 6. Conclusions

- 292 1. On a 26-image same-sequence holdout, the archived Mask R-CNN  
 293 checkpoint achieved union-mask IoU 0.652, Dice 0.769, and projected-  
 294 coverage MAE 0.0051. It outperformed fixed Otsu and training-only  
 295 Gaussian pixel baselines.
- 296 2. The split is not independent and instance AP is 0.249; the results do not  
 297 support individual-bubble or experiment-level generalization claims.
- 298 3. Complete 45 V sequences reproduce archived means to a four-case MAE  
 299 of 0.00085, but temporal autocorrelation reduces effective sample sizes

- 300 to 3.5–11.2 and materially widens mean intervals.
- 301 4. Projected coverage correlates with heat flux within each sweep, but
- 302 both follow voltage. Adding coverage does not improve cross-validated
- 303 prediction of mean measured surface temperature beyond heat flux,
- 304 mass flux, and inlet subcooling.
- 305 5. In a timestamp-registered 10 g/s screen, AE activity co-varies with the
- 306 imposed ramp while the terminal projected-coverage value decreases.
- 307 This is not trigger-synchronized or causal evidence.
- 308 6. The defensible output is a two-dimensional projected vapor-coverage
- 309 measurement plus a bounded multimodal screening workflow. Volu-
- 310 metric void fraction, universal regime thresholds, CHF detection, cal-
- 311 ibrated heat-transfer prediction, and acoustic inference require addi-
- 312 tional validation.

### 313 **CRedit authorship contribution statement**

314 Abrar Fahim: Data curation, Software, Formal analysis, Visualization,

315 Writing – review and editing. Farshad Barghi Golezani: Methodology, Soft-

316 ware, Visualization, Formal analysis, Writing – review and editing. Daniel

317 Curl: Investigation, Data curation, Methodology, Validation. Mohammad

318 Ishraq Hossain: Investigation, Data curation, Formal analysis, Writing –

319 original draft. Stephen Pierson: Contribution roles to be confirmed. Chirag

320 Kharangate: Contribution roles to be confirmed. Han Hu: Conceptualiza-

321 tion, Supervision, Funding acquisition, Project administration, Methodology,

322 Writing – review and editing. All roles require author confirmation before

323 submission.

### 324 **Declaration of competing interest**

325 The authors declare that they have no known competing financial in-

326 terests or personal relationships that could have appeared to influence the

327 work reported in this paper. All authors must confirm this statement before

328 submission.

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332 verified before submission.

### 333 **Data availability**

334 Analysis code is available at [https://github.com/UARK-NED3/](https://github.com/UARK-NED3/BubbleID-Flow)  
335 **BubbleID-Flow**. The Aug. 9 fine-tuned Mask R-CNN weights are available  
336 through the OSF view-only record at [https://osf.io/xnkh6/overview?](https://osf.io/xnkh6/overview?view_only=09d63d516ca0489e90fa9b94d37b529e)  
337 [view\\_only=09d63d516ca0489e90fa9b94d37b529e](https://osf.io/xnkh6/overview?view_only=09d63d516ca0489e90fa9b94d37b529e). The verified local  
338 checkpoint SHA256 is 6F60969CE876F57A78B53FC61895C99B92CB2E1B6F0  
339 E0B30F01FA301236042CD. Raw optical and thermal records are maintained  
340 in the project archive but are not publicly released in the manuscript  
341 package; access remains subject to sponsor, collaborator, and institutional  
342 data-sharing approval.

### 343 **Declaration of generative AI and AI-assisted technologies in the** 344 **manuscript preparation process**

345 During preparation of this work, the authors used OpenAI Codex to assist  
346 with manuscript drafting, code organization, analysis scripts, and determin-  
347 istic figure preparation. No generative image tool was used to create or alter  
348 the submitted scientific figures. The authors reviewed the resulting content  
349 and remain responsible for the manuscript.

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